DATA 266 — Homework 9

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SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

Notebook: `latent_diffusion.ipynb`. Dataset: FashionMNIST.

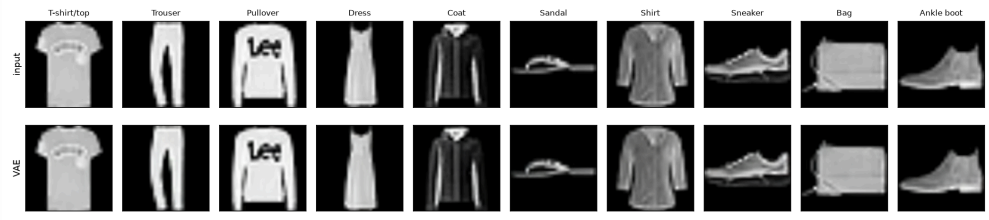
Same SID/SEED as the earlier homework. I did not train a second HP_ID network. The two models are the pretrained VAE and one conditional U-Net.

1. VAE compression

I loaded `stabilityai/sd-vae-ft-mse` and froze it. FashionMNIST is 28×28 grayscale. This VAE expects 3 channels and a size that divides by 8, and it downsamples by 8. I resized each image to 128×128 and repeated the gray channel, so the latent is 4×16×16. That is the map the U-Net attends over.

Diffusion uses the scaled latent, z × 0.18215. The decoder was trained on the unscaled latent, so I divide by 0.18215 before decoding. I used the posterior mean, so each image has one latent.

On all 10,000 test images, the MSE against the 128×128 encoder input, in [-1, 1], was **0.000781**.



Reconstructions

The top row is the resized input and the bottom row is the VAE decode, one test image per class. The copies are close. The mark on the pullover is still readable.

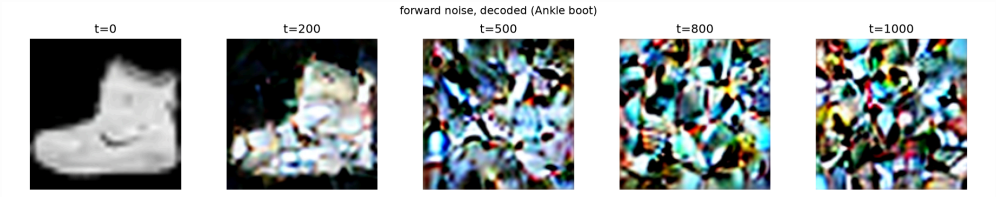
2. Forward diffusion in latent space

I used T = 1000. β goes linearly from 0.0001 to 0.02. The closed form is on the scaled latent:

[z_t = z_0 + , (0,I).]

The schedule has indices 0..999, so t=1000 in the figure is the last step, index 999. t=0 is the clean latent.

time	index	$\bar{\alpha}$
200	200	0.656347
500	500	0.077797
800	800	0.001508
1000	999	0.000040



Forward diffusion

The image is the first training sample, an ankle boot. At t=0 the boot is clear. At t=200 the shape is still there, with color noise on it. At t=500 the boot is mostly gone. At t=800 and t=1000 the decode is colored static. A noisy 4-channel latent does not come back as a gray FashionMNIST image.

3. U-Net and training

The denoiser takes a 4×16×16 latent. Base channels are 64 and the multipliers are (1, 2, 4), so the maps are 16×16, 8×8, and 4×4. Each level has residual blocks with GroupNorm and SiLU, plus skip connections. Self-attention is on the 16×16 maps and at the bottleneck.

Time is a sinusoidal embedding through an MLP. There are 10 class labels plus one null label. I add the label embedding to the time embedding. During training I replace the label with the null label with probability 0.1.

Setup

Item	Value
Dataset	FashionMNIST, resized to 128×128, 3 channels
VAE	`stabilityai/sd-vae-ft-mse`, frozen
Latent	4×16×16, posterior mean, times 0.18215
Device	MPS
Diffusion steps	1000
β start	0.0001
β end	0.02
Batch size	128
Epochs	50
Learning rate	2e-5
Optimizer	Adam
Label dropout	0.1, null label = 10
Guidance scale w	3
U-Net	base 64, multipliers (1, 2, 4), GroupNorm, SiLU, self-attention at 16×16 and the bottleneck
U-Net parameters	12,289,540
Model output shape	`(2, 4, 16, 16)`
Test reconstruction MSE	0.000781

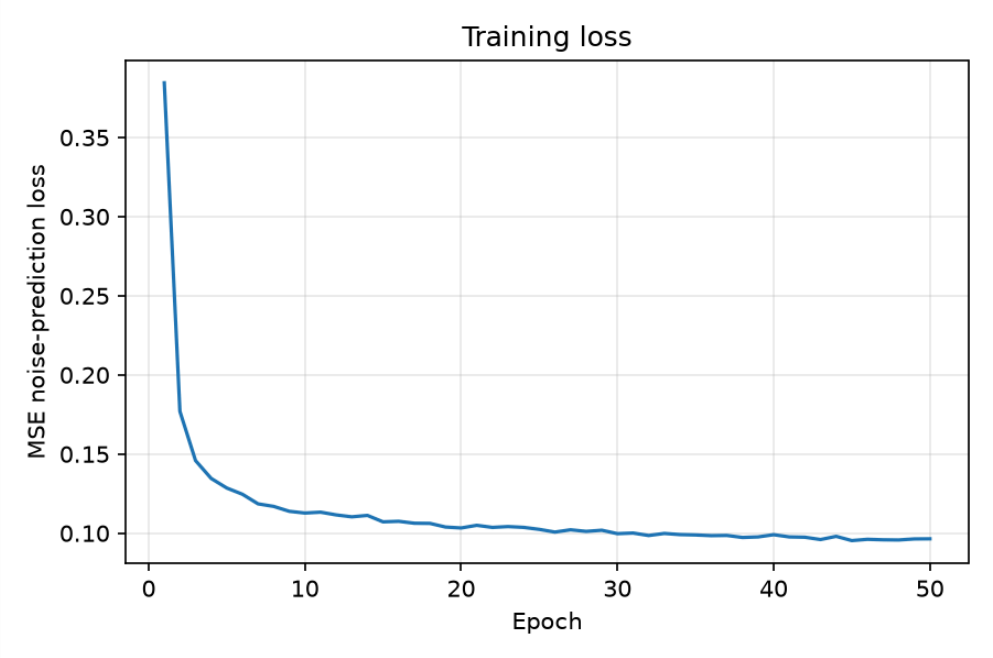
The model has **12,289,540 parameters**. The output check printed `torch.Size([2, 4, 16, 16])`.

I trained it to predict the Gaussian noise. Adam, learning rate 2e-5, batch size 128, 50 epochs, on MPS. That training took 111.44 minutes.

Training loss

Epoch	Noise-prediction MSE
1	0.384546
10	0.112722
20	0.103292
30	0.099690
40	0.099033
50	0.096441

The loss went from **0.384546** in epoch 1 to **0.096441** in epoch 50. It dropped quickly and then flattened near 0.096. A few later epochs moved by about 0.001, so the tail is flat but not perfectly smooth. All 50 epochs are in `RUN_LOG.txt`.



Training loss

4. Sampling

I ran the reverse chain for all 1000 steps. At each step I predict the noise with the class label and with the null label, and combine them as

[= (1+w)*{cond}* - *w*{uncond}]

with w = 3. The 64-image grid repeats classes 0–9 in row-major order. The last four labels are 0, 1, 2, and 3. That grid took 1.08 minutes.

Item	Value
Reverse steps	1000
Grid	64 images, classes 0–9 repeated row-major, w = 3
64-image sampling time	1.08 minutes



CFG samples

Most of the 64 cells look like clothing: shirts, trousers, dresses, shoes, and bags. Some coats, bags, and sandals are still glassy or mixed. I am not calling every cell a clean example of its label.

The comparison figure uses one starting noise and one image per class. The unconditional row does not follow the column label. Several columns are a different kind of clothing. The conditional row usually matches the label. With w=3 the items are simpler and closer to a typical piece of that class. On this draw the dress gets plainer and the sandal gets flatter. This is one noise tensor, so I would not claim more than that.



Guidance comparison

Conclusion

The VAE round trip was the clean part of the run. Test MSE was 0.000781, and the reconstructions stay close to the resized inputs. The forward process does what it should: the ankle boot fades, and by the last step $\bar{\alpha}$ is about 4e-5. The U-Net loss fell from 0.384546 to 0.096441. Classifier-free guidance with w=3 made the class match clearer on the comparison figure. The 64-image grid is recognizable clothing, with some distorted shoes and bags.